MIL-I-46894 (MI)
14 December 1973
SUPERSEDING
MIS-14141
20 October 1964

MILITARY SPECIFICATION

INSULATION PAPER, ASBESTOS CHRYSOLITE

This specification is approved for use by all activities of the Department of the Army.

1. SCOPE

1.1 Scope. This specification covers one grade of thermal insulating paper of organically bonded chrysolite asbestos fibers.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification, to the extent specified herein.

SPECIFICATIONS

Military

MIL-I-16688	Insulation Felt, Thermal, Fibrous Mineral (Semi-Rigid) (Light Weight)
MIL-I-45208	Inspection System Requirements

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing Materials.

ASTM D 149	Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Matrials at Commercial Power Frequencies, Test for

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania 19103.)

Consolidated Classification Committee

Consolidated Freight Classification Rules

(Application for copies should be addressed to Consolidated Classification Committee, 202 Union Station, Chicago 6, Illinois 60607.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample shall meet all the requirements of this specification.

3.2 Dimensions.

3.2.1 Thickness. The strips shall be furnished in thicknesses as specified in the contract or purchase order and shall conform to the requirements of Table I.

Table I

Permissible Limits in Specified Thickness

Specified Thickness, Inch	Min	Max
0.031	0.030	0.037
0.062	0.058	0.070
0.125	0.115	0.138

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3.2.2 Width. The strips shall be furnished in widths as specified in the contract or purchase order, with permissible variations of minus 1/16 inch and plus 1/8 inch.

3.2.3 Length. Unless otherwise specified, the strips shall be furnished in lengths, as specified in the contract or purchase order, with permissible variations of minus 1/8 inch and plus 1/4 inch.

3.3 Physical properties. The physical properties of the material, when conditioned in an atmosphere of 50 percent relative humidity at 22 degrees Celsius (C) (72 degrees Fahrenheit (F)) for a minimum of 4 hours, shall be as shown in the ap plication paragraph and Table II.

3.3.1 Basis weight. The basis weight or weight per 100 square feet of material shall conform to the limiting requirements of Table II.

3.3.2 Dielectric strength. The dielectric strength in volts per mil thickness shall not be less than the minimum requirements of Table II.

3.3.3 Hygroscopic moisture content. The hygroscopic content shall not exceed the maximum requirement of Table II.

Table II.
Physical Properties

Specified Thickness Inch	Basis Weight lbs/100 sq. ft. Min	Basis Weight lbs/100 sq. ft. Max	Dielectric Strength Volts/Mil, Min	Hygroscopic Moisture, % Max
1/32	15	20	450	1.25
1/16	30	35	250	1.25
1/8	60	70	150	1.25

Note: The value of these physical properties apply only to material conditioned in an atmosphere of 50 percent relative humidity at 22 degrees C (72 degrees F).

3.4 Chemical properties. The chemical composition of the finished material on a hygroscopic moisture free basis, shall be as shown in 3.4.1 through 3.4.4.

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3.4.1 Asbestos content. The content of chrysolite asbestos fibers shall be not less than 98 percent by weight of the finished material.

3.4.2 Total organic content. The total organic content shall be not greater than 1.5 percent by weight of the finished material.

3.4.3 Total iron content. The total iron content, as iron oxide (Fe_2O_3) shall be not greater than 6 percent by weight of the asbestos fiber content of 3.4.1.

3.4.4 Thermal conductivity. The thermal conductivity in BTU-In/Hr/Sq Ft/Degrees F of the finished material, in the as received condition, shall not exceed the mean temperature values of Table III.

Table III.

Thermal Conductivity	
Mean Temperature °F	Thermal Conductivity BTU-In/Hr/Sq Ft/°F
250	0.85
300	0.88
350	0.91

3.5 Workmanship. The material shall be uniform in quality, clean and free from defects detrimental to fabrication, appearance or performance of the insulation paper.

4. QUALITY ASSURANCE

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent production lots of asbestos chrysolite insulation paper.

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The preproduction sample shall be subjected to all examinations and tests specified herein. Unless otherwise specified, the Government will perform the examinations and tests for preproduction sample at the contractor's plant. Preproduction samples which do not meet all requirements of this specification shall be rejected and returned to the contractor. Subsequent quantities will not be considered for acceptance until approval of the preproduction sample has been obtained.

4.3 Inspection provisions.

4.3.1 Lot size. A lot shall consist of not more than 25,000 square feet of finished material of the same thickness and basic weight from one production run submitted for inspection and acceptance at one time.

4.3.2 Sampling.

4.3.2.1 For visual and dimensional examination. Sampling for the visual and dimensional requirements of 3.2, 3.5, and 5 shall be in accordance with MIL-STD-105 and Table IV herein.

Table IV.

Sampling Plan

Number of Units In Inspection Lot	Number of Units In Inspection Sample	Acceptance Number (Defective units)	Rejection Number (Defective units)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	1	2
111 to 300	25	1	2
301 and over	35	2	3

Note: A unit shall consist of the number of strips in a package containing 50 pounds of strips. If the inspection lot contains less than seven units, each unit shall be inspected.

4.3.2.2 For examination testing.

4.3.2.2.1 Sampling for lot acceptance tests. From each inspection sample the number of strips shown in Table V shall be selected at random and submitted to the tests specified in 4.3.2.2.

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Table V.

Sampling Plan for Lot Acceptance Tests	
Number of Units in Inspection Sample	Number of Strips to be Selected for Lot Acceptance Tests
7	5
10	5
15	10
25	10
35	15

4.3.2.2.2 Sampling for thermal conductivity test. When specified in the contract or purchase order, the contractor shall furnish with the lot two specimens, or equivalent amount of material, 1 inch thick and 8 inches in diameter for the thermal conductivity test. The specimens or material so selected shall be suitably marked and dated for identification with the lots represented.

4.3.3 Examination.

4.3.3.1 Visual examination. Visual examination of samples specified in 4.3.2.1 shall be conducted for the purpose of determining compliance with the requirements specified in 3.2 and 3.5.

4.3.3.2 Examination testing. Examination testing of the samples specified in 4.3.2.2 to determine compliance with the following characteristics, shall be conducted in accordance with their corresponding test paragraphs.

a. Basis weight, 4.4.1.

b. Dielectric strength, 4.4.2.

c. Hygroscopic moisture, 4.4.3.

d. Asbestos content, 4.4.4.

e. Total organic content, 4.4.5.

f. Total iron content, 4.4.6.

g. Thermal conductivity, 4.4.7.

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4.4 Test methods and procedures.

4.4.1 Basis weight.

4.4.1.1 Apparatus. The apparatus used for the basis weight test shall be a balance or scale having an accuracy of plus or minus 0.25 percent of the applied load.

4.4.1.2 Test. From each strip in the test sample of 4.3.2.2.1 a specimen shall be cut 10 inches long to include the full width of the strip and the area shall be determined. Each specimen shall then be conditioned in an atmosphere of 22 degrees C (72 degrees F) for at least 4 hours. The conditioned specimens shall then be combined and weighed. The basis weight for the inspection lot shall then be calculated and reported as the weight in pounds for 100 square feet of material.

Calculation:

$$\text{Weight of 100 sq. ft.} = \frac{B \times C}{A}$$

Where: A = area of specimen
B = weight of specimen
C = 100

Calculated weight shall conform to Table II.

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4.4.2 Dielectric strength.

4.4.2.1 Apparatus. The apparatus used for the dielectric strength test shall be as required in ASTM D 149.

4.4.2.2 Test. From each strip in the test sample 4.3.2.2.1 three specimens 2.5 inches in length shall be cut to determine the dielectric strength of the material. Each specimen shall be conditioned in accordance with the procedured described in 4.4.1 and tested in accordance with ASTM D 149. Test results shall then be reported as minimum, maximum and average volts per mil. The measured dielectric strength shall conform to Table II.

4.4.3 Hygroscopic moisture.

4.4.3.1 Apparatus. The apparatus used for the hygroscopic moisture test shall consist of the following:

a. Platinum or porcelain crucible suitable for use at 105 degrees C (221 degrees F).

b. Analytical balance having an accuracy of plus or minus 0.25 percent of the applied load.

4.4.3.2 Test. From each strip in the test sample of 4.3.2.2.1 three specimens approximately 2 inches in length shall be cut and conditioned in accordance with the procedure described in 4.4.1. Each conditioned specimen shall then be placed in a desiccated crucible, weighed on an analytical balance, heated at 105 degrees C (221 degrees F) for 2 hours, and reweighed. The average percent loss in weight of the specimen shall be reported as the hygroscopic moisture content, and shall be as specified in Table II.

Calculation:

$$\text{Percent hygroscopic moisture content} = \frac{A-B}{A} \times 100$$

Where: A = unheated specimen
B = heated specimen

4.4.4 Asbestos content.

4.4.4.1 Apparatus. The apparatus used for the asbestos content test shall consist of the following:

a. Analytical balance having an accuracy of plus or minus 0.25 percent of the applied load.

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b. Crucible suitable for use at 900 degrees C (1652 degrees F).

4.4.4.2 Test. Heat each specimen and crucible to 871 to 900 degrees C (1600 to 1652 degrees F) for 1 hour to burn off the organic binder, desiccate for a minimum of 2 hours, and weigh. The weight of the ash remaining in each crucible shall then be determined. The minimum percentage of asbestos content shall be calculated and reported on a moisture free basis, using the final specimen weight, and shall be as specified in 3.4.1.

Calculation:

$$\text{Percent asbestos} = \frac{A \times 100}{B \times 0.86}$$

Where: A = weight of ash
B = weight of specimen

4.4.5 Total organic content.

4.4.5.1 Apparatus. The apparatus used for the total organic content test shall consist of the following:

a. Analytical balance having an accuracy of plus or minus 0.25 percent of the applied load.

b. Crucible suitable for use at 900 degrees C (1652 degrees F).

4.4.5.2 Test. The total organic content shall be calculated, using the procedure in 4.4.5.2 as the difference of 100 percent minus percent of asbestos. The results reported shall include maximum and average percentage of the total organic content, and shall be as specified in 3.4.2.

Calculation:

Percent organic content = A - B

Where: A = 100 percent
B = percent asbestos

4.4.6 Total iron content.

4.4.6.1 Reagents. The reagents used for the total iron content shall consist of the following:

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a. Hydrocloric acid (specific gravity 1.19).

b. Hydrofluoric acid (chemically pure).

c. Distilled water.

d. Sulfuric acid (specific gravity 1.84).

e. Phenanthroline indicator.

f. Ceric-ammonium sulfate.

4.4.6.2 Apparatus. The apparatus used for the total iron content shall consist of the following:

a. Glass beaker of 300 milliliter (ml) capacity.

b. Walden silver reductor, or approved equal.

4.4.6.3 Test. From each specimen in 4.4.4 an aliquot portion of a minimum of 1 gram (g) of ash shall be transferred to a 300 ml glass beaker, and 25 ml of concentrated hydrochloric acid added. This mixture shall be boiled and during the boiling, chemically pure hydrofluoric acid shall be added by dropping until a volume of 15 ml is reached. The solution shall then be cooled and then diluted with 100 ml of distilled water. The solution shall then be passed through a Walden silver reductor or approved equal and the reductor washed with 100 ml of dilute hydrochloric acid (ratio 1:10). To the resulting filtrate shall be added a solution of 15 to 20 ml of concentrated sulfuric acid (sp gr 1.84) and three drops of phenanthroline indicator and the solution then titrated with a standardized solution of ceric-ammonium sulfate. The weight of the iron, thus determined for each specimen, shall be calculated and reported as the percentage of iron oxide (Fe_2O_3) in the specimen on a moisture free basis. The iron content shall be as specified in 3.4.3.

Calculation:

$$\text{Percent } Fe_2O_3) = \frac{1.428 \times A \times B \times 0.05585}{C} \times \frac{D}{E} \times \frac{100}{1}$$

Where: A = milliliters, ceric-ammonium sulfate
B = normality, ceric-ammonium sulfate
C = weight of specimen
D = weight of ash
E = weight of aliquot

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4.4.7 Thermal conductivity.

4.4.7.1 Apparatus. The apparatus used for the thermal conductivity test shall be as required in MIL-I-16688.

4.4.7.2 Test. The thermal conductivity of the specimens furnished in 4.3.2.2 shall be determined in accordance with the guarded hot plate method described in MIL-I-16688. The two specimens shall be equally balanced for actual density and thickness prior to performing the test. The result shall be reported as: BTU-In/Hr/Sq Ft/°F. The conductivity shall be as specified in 3.4.4.

4.4.8 Rejection. Individual units or lots, as applicable, of asbestos sheets which do not meet the requirements of this specification shall be rejected and returned to the contractor.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level C as specified.

5.1.1 Level C. Insulation paper shall be packaged in accordance with the best commercial practices.

5.2 Packing. Packing shall be Level C as specified.

5.2.2 Level C. Insulation paper shall be packed in a manner which will afford adequate protection against damage during shipment from supply source to the first receiving activity. Container shall conform to applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking specified in the contract or order (see 6.2) each container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The chrysolite asbestos insulation paper covered by this specification is used in components requiring an inexpensive supplementary thermal barrier for applications up to 500 degrees C (932 degrees F).

6.2 Ordinary data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required.

c. Where the preproduction sample test shall be performed.

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d. Sampling if other than specified (see 4.3.2).

e. Length, width, and thickness of strip required (see 3.2)

f. Total quantity desired.

g. Any special markings required.

h. Thermal conductivity specimens, if required (see 4.3.2.2.2).

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-14141, dated 20 October 1964.

Custodian:
ARMY-MI

Preparing Activity:
ARMY-MI
Project No. 5640-A024

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPPETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO (If "yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional) | DATE

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